Kashish E-Commerce Sales Analytics

Project Objective:

The primary objective of this project is to analyze the online sales data of **Kashish E-Commerce Store** and develop an interactive Power BI dashboard that provides meaningful business insights. The dashboard enables stakeholders to monitor sales performance, identify profitable products, understand customer purchasing behavior, and support data-driven decision-making across different regions of India.

Business Problem Statement:

- Calculate key business metrics such as total sales, total profit, total quantity sold, and average order value.
- Analyze sales performance across different states in India.
- Identify the most profitable product categories and sub-categories.
- Evaluate monthly profit trends and seasonal business performance.
- Understand customer payment preferences.
- Provide an interactive dashboard for business monitoring using dynamic filters.

Data Cleaning &Transformation:

Power Query was used to prepare the dataset before analysis.

- Imported and validated multiple CSV datasets.
- Checked for missing and inconsistent values.
- Corrected data types for numerical and date columns.
- Removed unnecessary columns.
- Standardized category and payment mode values.
- Created relationships between datasets using Order ID.

Measures:

- Total Sales
- Total Profit
- Total Quantity Sold
- Average Order Value

Dashboard Development:

The dashboard was designed with a Business Intelligence approach it includes:

- KPI Cards
- Profit by Sub-Category Analysis

- State-wise Sales Analysis
- Monthly Profit Trend
- Category-wise Quantity Distribution
- Payment Mode Analysis
- Quarter and State Slicers for interactive filtering

Key Business Insight:

- The dashboard highlights several important findings:
- Total Sales exceeded ₹438K.
- Total Profit reached approximately ₹37K.
- More than 5,600 products were sold.
- Average Order Value remained around ₹121K.
- Maharashtra generated the highest sales among all states.
- Printers emerged as the most profitable product sub-category.
- Clothing contributed the highest quantity sold.
- Cash on Delivery (COD) was the most preferred payment method.
- Monthly profit analysis revealed seasonal fluctuations, including a few low-profit periods.

Business Impact:

The dashboard provides business users with a centralized reporting solution capable of:

- Monitoring overall business performance.
- Identifying top-performing products and categories.
- Understanding customer purchasing behavior.
- Tracking regional sales performance.
- Supporting faster, data-driven business decisions.
- Improving sales strategy through actionable insights.

Skill Demonstrated:

- Microsoft Power BI
- Power Query
- Data Modeling
- Data Cleaning
- Data Transformation
- Interactive Dashboard Design
- KPI Development
- Data Visualization
- Business Intelligence
- Sales Analytics
- Business Storytelling

Challenges Faced:

- Cleaning and validating transactional sales data.
- Building relationships between multiple datasets.
- Designing an intuitive and responsive dashboard layout.
- Ensuring interactive filtering across all visualizations.
- Balancing dashboard aesthetics with readability and performance.

Future Improvements:

- Add geographic map visualizations for regional performance.
- Enhance the dashboard with advanced Power BI features such as bookmarks and tooltips.

Conclusion:

This project demonstrates how Microsoft Power BI can be used as a complete Business Intelligence platform when combined with Power Query, Power Pivot, and interactive dashboard reporting.

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